

MAB Pioneer (Coastal Virginia) & ROMS Model–Observation Comparison Research Constitution

Draft — 2026-04-14

Research Context

This project compares observations from the OOI Pioneer MAB South (Coastal Virginia) Array with regional ocean model output (e.g., DOPPIO / ESPreSSO / NYB-ROMS) during a **known hurricane event** that passed over the array. The goal is to evaluate how both the moored observations and the ROMS simulation capture the event – its timing, magnitude, vertical structure, and recovery – and to identify where model and observations diverge.

Event-scale comparison (rather than seasonal-climatology comparison) is the primary framing: hurricane passage is a well-defined forcing that stresses both the observing array and the model, so mismatches are diagnostic rather than statistical.

The specific hurricane, time window, and metrics live in the project spec. This constitution governs the standards that apply regardless of the specific question framing.

Core Principles

I. Reproducibility

Every result traces to version-controlled code, pinned environments (`pyproject.toml` / `uv.lock`), and documented data snapshots. Raw model output and observations are downloaded by scripts with logged URLs, checksums, and retrieval dates – not hand-curated.

II. Data Integrity

Raw ROMS output and OOI NetCDF files are immutable. All regridding, interpolation, and subsetting produces new files in `outputs/data/`, never overwrites sources. Missing or masked values are preserved as NaN with a flag column, not silently dropped.

III. Provenance

Every figure, table, and statistic links back to: the code that produced it, the input files (including model run ID and deployment number), and the coregistration method used. Model-observation comparison plots must identify both datasets explicitly in the caption (product name, version, time range, vertical level or depth).

IV. Coregistration Discipline

Model and observation comparisons require explicit decisions about: - **Time**: matching window, temporal averaging (hourly/daily/monthly), lag treatment. Default is to average both products to the coarser native cadence rather than interpolate the finer one. - **Space**: the model grid cell(s) used for a given mooring location, how horizontal interpolation was done (nearest / bilinear), and how land masking / wet-point checks were handled. - **Depth**: how model sigma levels were mapped to observation depths (e.g., linear in z , pressure-based). State assumptions explicitly. - **Variable identity**: potential vs. in-situ temperature, practical vs. absolute salinity, pressure vs. depth. Convert before comparing; do not assume equivalence.

V. Model Humility

ROMS output is a simulation, not truth. Observation-model differences are not automatically model errors – they can reflect observation representativeness, unresolved processes, spin-up, open-boundary forcing, or simply a different realization of turbulence. Discuss mismatches as hypotheses to investigate, not as corrections to apply to the model.

VI. Honest Framing

State what this project can and cannot conclude. A single mooring evaluated against a regional model constrains some statistics (seasonal cycle, mean stratification, event response) and cannot constrain others (mesoscale eddy statistics, basin-wide heat budgets). Make those boundaries explicit in abstracts, captions, and READMEs.

Data Sources

00I Pioneer MAB South / Coastal Virginia Array (2024–present)

- **Description**: 00I Coastal & Global Scale Node array relocated from New England to the Mid-Atlantic Bight south of the Chesapeake Bay shelf

break (~37°N, 75°W region). Surface moorings, profiler moorings, and gliders spanning inner shelf to slope.

- **Access:** OOI Data Explorer (<https://dataexplorer.oceanobservatories.org/>), OOI Net / M2M API, and – where mirrored – `/home/jovyan/ooi/kdata/`.
- **Temporal coverage:** Deployments starting 2024; treat as an actively growing record.
- **Key variables:** temperature, salinity, pressure, velocity, optical backscatter, fluorescence (platform-dependent).
- **Known issues:** New array – telemetered vs. recovered stream availability varies; deployment-to-deployment depth coverage must be audited (per CP02PMU0 lesson); QARTOD flags still maturing for some instruments.
- **Documentation:** <https://oceanobservatories.org/array/coastal-pioneer-mab/>

DOPPIO (Rutgers MAB/GoM ROMS) – primary model

- **Description:** Rutgers Ocean Modeling Group’s data-assimilative ROMS simulation covering the Mid-Atlantic Bight and Gulf of Maine. The authoritative regional ROMS product for this project.
- **Access:** THREDDS/OPeNDAP at <https://tds.marine.rutgers.edu/> (document the exact dataset URL, version, and retrieval date in `outputs/data/README.md` and in any downloader script).
- **Known issues:** Spin-up periods, open-boundary artifacts near domain edges, vertical sigma coordinate must be converted to `z` for comparison, land mask vs. wet-cell mismatches, assimilation increments can produce timing-specific features.

Secondary ROMS products (for cross-check only)

- **ESPreSSO** – older MAB ROMS; useful for a second-opinion check but not the anchor.
- **NYB-ROMS** – smaller New York Bight domain; include only if a given version covers the Pioneer MAB South footprint.
- Do not swap the primary-vs-secondary role without updating the spec.

Technical Environment

- **Language:** Python ≥3.12 (JupyterHub environment)
- **Package management:** `uv` (use `uv add`, not `pip install`)

- **Key packages:** xarray, pandas, numpy, netCDF4, matplotlib (publication), plotly (exploratory), and ROMS-specific helpers (xroms or equivalent) added as the work requires them.
- **Compute environment:** JupyterHub container (CPU-capped; 24-worker ceiling for parallel jobs per user's global rule).
- **Version control:** Git. Raw data stays outside the repo; processed slices in outputs/data/ (large files gitignored).
- **Self-contained notebooks:** Every notebook declares and installs its own dependencies in the first code cell (e.g., `%pip install xarray netCDF4 matplotlib plotly` or equivalent) so it runs standalone in a fresh Jupyter / Colab environment without relying on the project's uv venv. Pin versions in the install cell. The project pyproject.toml remains the authoritative environment for scripts and tests, but notebooks must not assume it.

Coordinate Systems & Units

- **Spatial reference:** WGS84 (EPSG:4326) for lat/lon. ROMS grids are curvilinear – document which horizontal interpolation method is used to get to observation locations.
- **Vertical:** ROMS native sigma; convert to depth (m) or pressure (dbar) explicitly before comparison.
- **Time zone:** UTC.
- **Standard units:** °C, dbar, PSU (practical salinity), m/s.
- **Missing data:** NaN.

Figure Standards

Primary audience is **students** working through the notebooks as a teaching demo, so figures default to interactive and exploratory.

- **Default (student-facing notebooks):** Plotly – interactive, hover for values, zoom/pan. Every figure should invite exploration.
- **Optional publication export:** If a figure is promoted to a paper or static deliverable, re-render in matplotlib at 300 dpi PNG or PDF vector.
- **Colormaps:** Colorblind-safe (viridis , cividis , cmoccean.thermal , Okabe-Ito for categorical).

- **Comparison plots:** Always label which product is which; use consistent color assignments across figures (e.g., observations in black, DOPPIO in a sequential color).
- **Rubrics:** Score every figure against `timeseries-figure` or `map-figure` rubric before committing. Student-lab tier: clarity and pedagogy first; paper tier only when exporting a final figure.
- **Captions:** Write for a student reader – say what the figure shows and why it matters, not just the axes.

Quality Checks

- **Observations:**
 - QARTOD flags: accept only flag=1 (pass) or 2 (not evaluated).
 - Range checks (typical MAB shelf/slope bounds):
 - Temperature: 2–28 °C
 - Salinity: 28–37 PSU
 - Pressure: 0–500 dbar
- **Hurricane-event excursions:** The bounds above are intentionally set for typical conditions. Real storm effects (rapid surface cooling from mixing, freshening from rain/runoff, surge at the surface mooring) can legitimately push values outside these bounds. Flagged excursions during the event window must be **reviewed case-by-case**, not silently accepted or rejected. This is a teaching feature: students see the QC system fire and investigate whether the flag reflects an instrument problem or a real oceanographic signal.
- **Model:**
 - Respect the land/wet mask; do not compare at masked cells.
 - Exclude documented spin-up window and open-boundary buffer cells.
- **Comparison:**
 - Match units and variable identity before taking differences.
 - Report sample size for every comparison statistic.

AI Authorship & Disclosure

All AI-generated prose (methods text, notebook markdown cells, READMEs, figure captions, report drafts) carries the standard italic disclosure label at the top and renders in Courier New monospace per the user's global

AI-disclosure rule. Code, code comments, commit messages, filenames, and short inline labels are exempt.

Testing

- `pytest` is mandatory. All non-trivial logic (loaders, regridders, coregistration helpers, metric calculators) has tests in `tests/`.
- Code must pass `uv run pytest` before every commit. Do not skip or disable failing tests.

Literature Handling

- Papers live under `literature/`
`{open_access,library_subscription,author_copies,embargoed}/` with a README inventory. Targeted `.gitignore`, not blanket `*.pdf`.
- Publisher PDFs (paywalled / author copies) are uploaded by the user. Claude does not curl-download outside fully open-access sources (CC-BY, PMC, public domain).

Project Notes

- **Repository visibility:** Public GitHub repo (`daxsoule/MAB_PioneerV_ROMS`).
- **Primary audience:** Students using the notebooks as a teaching demo. READMEs, notebook prose, captions, and choice of exploratory vs. static figures should all serve that audience first. Research outputs (figures, tables) can be promoted later if the work becomes publication-ready.
- **Status:** New project as of 2026-04-14. No publication timeline or embargo yet.
- **Focal event:** A 2025 hurricane passage over the Pioneer MAB South array. Storm name, track, and exact event window TBD in the spec.